

Machine Learning: The No-Nonsense (But Fun) Guide

Day 13: Gradient Boosting – The "Fix Your Mistakes with Math" Algorithm

1. Story & Intuition

Yesterday, AdaBoost focused on which samples were classified incorrectly and increased their importance.

Gradient Boosting asks a different question:

"How wrong was my prediction? By exactly how much? Let's learn that error and correct it."

Example: House Price Prediction

House	Actual Price	Predicted Price	Residual (Error)
A	\$500k	\$450k	+\$50k
B	\$300k	\$350k	-\$50k
C	\$700k	\$650k	+\$50k

Instead of predicting the original house prices again, Gradient Boosting trains a new model to predict these errors (residuals).

The corrected prediction becomes closer to the actual value after every iteration.

"AdaBoost says: 'You got this sample wrong.' Gradient Boosting says: 'You were off by \$50k—let me fix exactly that amount.'"

2. What is Gradient Boosting?

Gradient Boosting is an ensemble learning algorithm that builds models sequentially by minimizing prediction errors.

It combines:

- Additive Modeling
- Gradient Descent Optimization

Each new Decision Tree learns the residual errors made by the previous ensemble.

AdaBoost vs Gradient Boosting

AdaBoost	Gradient Boosting
Reweights samples	Fits residual errors
Exponential loss	Any differentiable loss function
Decision Stumps	Small Decision Trees
Adaptive weighting	Gradient Descent in function space

Core Idea

Instead of predicting the target directly,

Each new tree predicts:

Residual = Actual – Previous Prediction

The final prediction is the sum of predictions from all trees.

3. Working of Gradient Boosting

Step 1 – Start with a Simple Prediction

For regression problems, the first prediction is usually the mean of the target variable.

Example:

$\text{Prediction}_0 = \text{Mean}(\text{Target})$

$\text{Prediction}_0 = \$500,000$

Initially, every sample receives the same prediction.

Step 2 – Calculate Residuals

Residuals represent the prediction errors.

$\text{Residual} = \text{Actual Value} - \text{Predicted Value}$

Example

House	Actual	Prediction	Residual
A	\$500k	\$500k	\$0
B	\$300k	\$500k	-\$200k
C	\$700k	\$500k	+\$200k

These residuals become the target values for the next tree.

Step 3 – Train a Decision Tree on Residuals

Instead of learning the original values,

Tree 1 learns

House A $\rightarrow$ 0

House B $\rightarrow$ -200k

House C $\rightarrow$ +200k

The tree now understands how much correction is needed.

Step 4 – Update Predictions

The prediction is updated using the learning rate.

Formula

New Prediction

= Previous Prediction

+ Learning Rate $\times$ Tree Prediction

Example

Learning Rate = 0.1

House	Updated Prediction
A	\$500k
B	\$480k
C	\$520k

The predictions become closer to the actual values.

Step 5 – Repeat

The process repeats multiple times.

Each iteration

- Computes new residuals
- Trains another Decision Tree
- Updates predictions

Example

Round 0

Prediction = Mean

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Round 1

Prediction = Mean + $0.1 \times \text{Tree}_1$

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Round 2

$$\text{Prediction} = \text{Mean} + 0.1 \times \text{Tree}_1 + 0.1 \times \text{Tree}_2$$

↓

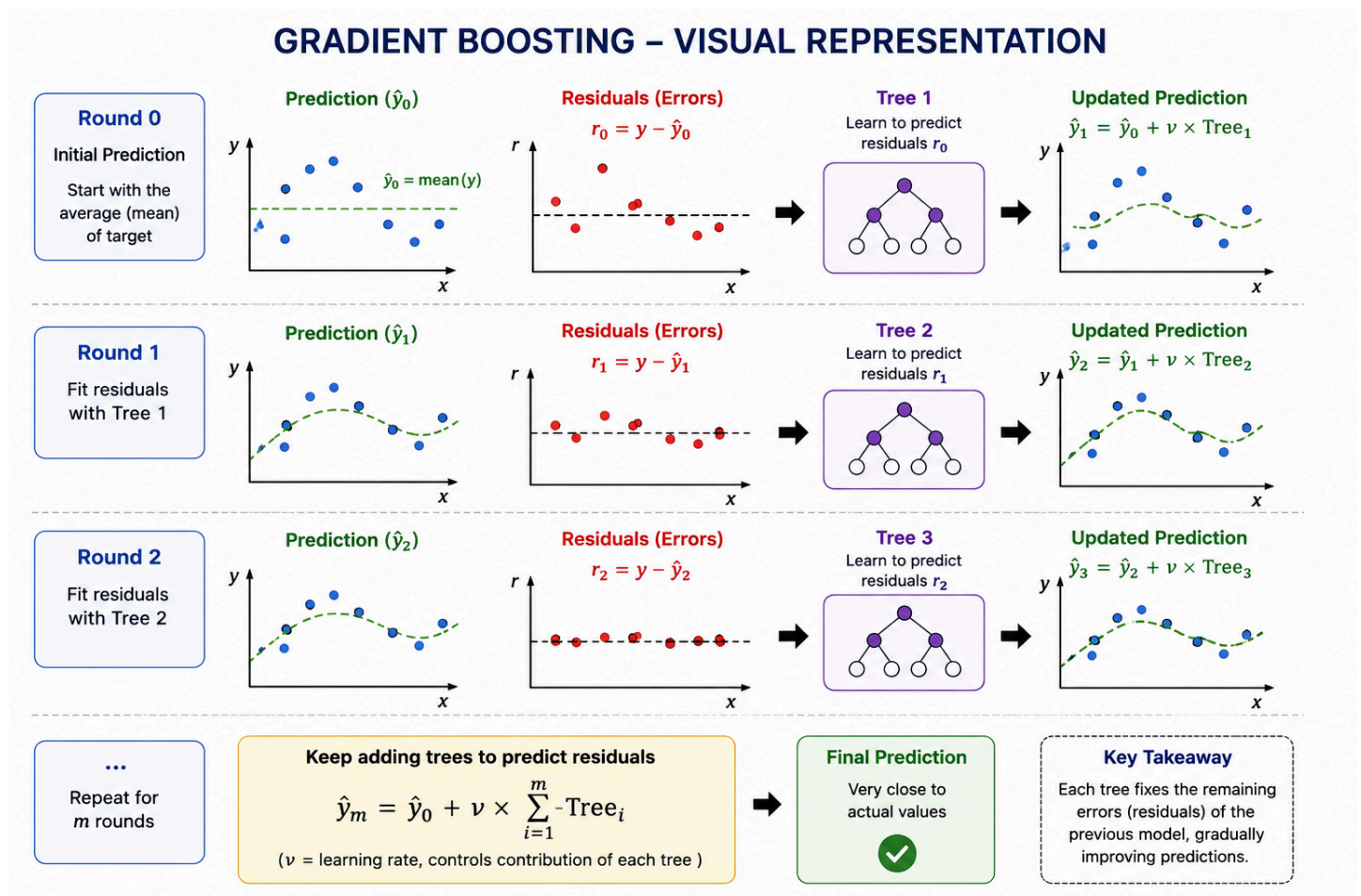
Round 3

$$\text{Prediction} = \text{Mean} + 0.1 \times \text{Tree}_1 + 0.1 \times \text{Tree}_2 + 0.1 \times \text{Tree}_3$$

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Continue until residuals become very small.

4. Visual Representation



5. Important Hyperparameters

Hyperparameter	Description	Recommended Value
n_estimators	Number of boosting rounds	100–1000
learning_rate	Contribution of each tree	0.01–0.30
max_depth	Maximum depth of each tree	3–6
min_samples_split	Minimum samples required for split	2–10
min_samples_leaf	Minimum samples in leaf	1–5
subsample	Fraction of training data used per tree	0.5–1.0
max_features	Features used for splitting	sqrt or 0.8

Golden Rule

Learning Rate × Number of Estimators ≈ Constant

Learning Rate	Estimators	Purpose
0.30	100	Fast baseline
0.10	300	Better accuracy

0.01	1000	Highest accuracy but slower
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6. Gradient Boosting vs AdaBoost vs Random Forest (which we have learnt)

Feature	AdaBoost	Gradient Boosting	Random Forest
Main Focus	Misclassified Samples	Residual Errors	Variance Reduction
Loss Function	Exponential	Differentiable Loss	Gini / Entropy
Tree Type	Decision Stumps	Small Trees	Deep Trees
Learning Style	Sequential	Sequential	Parallel
Learning Rate	Fixed	Adjustable	Not Required
Training Speed	Medium	Slower	Faster
Best For	Simple Problems	High Accuracy Tabular Data	General Purpose

7. Stochastic Gradient Boosting

Standard Gradient Boosting may overfit the training data.

To reduce overfitting, randomness is introduced.

Techniques

Technique	Purpose
Subsample	Randomly selects a portion of training data for each tree
Max Features	Randomly selects features at every split
Both Combined	Improves generalization and reduces variance

8. When to Use Gradient Boosting

Suitable For

- Structured tabular datasets
- High prediction accuracy
- Feature importance analysis
- Kaggle competitions
- Complex nonlinear relationships

Avoid When

- Image data
 - Text data
 - Very large datasets requiring fast training
 - Simple interpretable models
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9. Common Problems and Solutions

Problem	Cause	Solution
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Overfitting	Trees become too complex	Reduce max_depth, lower learning_rate
Underfitting	Model too simple	Increase estimators or tree depth
Slow Training	Sequential learning	Reduce estimators or use XGBoost
Unstable Results	High variance	Increase regularization and minimum split size

10. Real-World Applications

- Credit Risk Prediction
 - Demand Forecasting
 - Click-Through Rate Prediction
 - Medical Diagnosis
 - Kaggle Machine Learning Competitions
 - Customer Analytics
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11. Interview Questions

1. What is Gradient Boosting?
2. What are residuals?
3. How does Gradient Boosting differ from AdaBoost?
4. Why are shallow Decision Trees used?
5. Explain the role of the Learning Rate.
6. Why is Gradient Boosting called additive modeling?
7. How does subsampling reduce overfitting?
8. When would you choose Random Forest over Gradient Boosting?
9. How can overfitting be prevented in Gradient Boosting?
10. What is a staged prediction?

Happy Learning !!!!